AutoSteer-H: Dynamic Conditional Inference-Time Steering to Mitigate VLM Hallucinations

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Abstract

Vision-Language Models (VLMs) experience significant hallucinations, often due to misalignments between visual inputs and textual outputs [3]. Recent inference-time interventions aim to mitigate this by shifting latent representations globally, which also degrades performance on factual, benign queries. We propose **AutoSteer-H**, a framework that adaptively reduces hallucinations during inference without requiring tuning. By introducing a Truth Awareness Score (TAS) to pinpoint the intermediate layer most sensitive to factuality, we train a lightweight multi-layer perceptron (MLP) prober to predict the likelihood of hallucinations. A conditional steering head then activates dynamically to correct output only when a hallucination is likely. This method intends to cut down on hallucinations by ensuring visual grounding while maintaining the overall usefulness of the VLM.

1 Problem Statement

Large Vision-Language Models (LVLMs) face difficulties with object and attribute hallucinations due to misalignments between pre-trained vision encoders and text decoders that rely heavily on parametric memory[3]. Correcting these shortcomings via full finetuning, Reinforcement Learning from Human Feedback (RLHF), or instruction tuning is computationally expensive. This calls for an approach that is computationally efficient, effective in mitigating hallucinations and applied conditionally.

2 Background

Recent test-time interventions apply static, unconditional shifts across all layers to reduce hallucinations [3]. However, this "always-on" filtering demotes general capabilities on factual queries. We propose AutoSteer-H, an adaptive mechanism motivated by recent hate speech mitigation [1]. It identifies the most factuality-sensitive layer, finds the hallucination probability, and applies a dynamic correction vector *only* when necessary.

3 Related Work and Limitations

Editing Representations: Recent works project internal VLM representations to language vocabularies to identify hallucination root causes. They use targeted edits (e.g., orthogonalization) to force visual grounding [2]. But simple edits like orthogonalization might not come in handy to mitigate hallucinations with complex multi-token attention.

Latent Space Steering: Adding Principal Component Analysis (PCA)-derived vectors to representations provides a robust global solution [3], but this diminishes fine-grained, input-dependent control and neagtively affects factual generation.

Adaptive Steering: AutoSteer introduced a conditional mechanism for toxicity using a *Safety Awareness Score* to dynamically trigger a refusal head [1]. AutoSteer-H applies this efficient architecture to the factuality realm with a trainable hallucination probing network.

Inference-Time Interventions: PAI [4] highlights that hallucinations originate from "text inertia" and adaptively process through attention weights assigned to image tokens during decoding to force visual grounding. Similarly,

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AutoSteer-H utilizes training-free, dynamic interventions but specifically performs conditional latent steering. The only training involved in *AutoSteer-H* would be the prober network trained on the feature maps of the most layer chosen based on Truth Awareness Score, To classify the feature map as hallucination probable and factual probable.

4 Proposed Methodology

AutoSteer-H operates entirely at inference time without fine-tuning the base VLM. The pipeline consists of three lightweight stages:

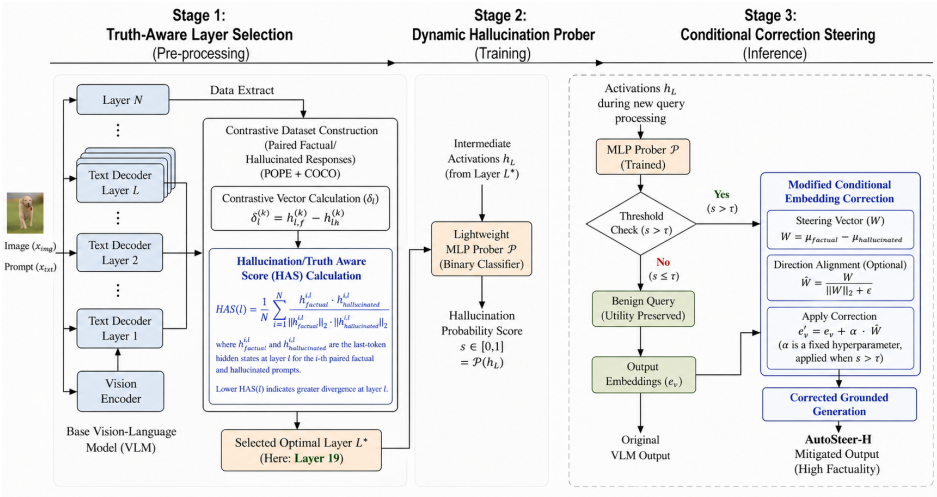

Figure 1: Workflow of the AutoSteer-H pipeline. A factuality-sensitive layer L is chosen, an MLP prober $\mathcal{P}$ is trained with input as the feature maps from layer L. At inference, if a hallucination is probable, steering is dynamically activated.

4.1 Truth-Aware Layer Selection

We construct a contrastive dataset of paired factual and hallucinated VLM responses for the same images. We extract the activation vectors $h_l(x_{\text{factual}})$ and $h_l(x_{\text{hallucinated}})$ at each intermediate layer l. We compute the contrastive vector δ_l and calculate the Truth Awareness Score (TAS) to find the layer with the highest score of factual distinctions capability:

$$TAS(l) = \frac{1}{N}\sum_{i=1}^{N} \frac{h_{i,l}^{factual}.h_{i,l}^{hallucinated}}{||h_{i,l}^{factual}||_2.||h_{i,l}^{hallucinated}||_2} \tag{1}$$

The layer l^* with the highest TAS is selected for probing.

4.2 Dynamic Hallucination Prober

Using layer l^*'s feature maps, we train a lightweight MLP prober $\mathcal{P}$ to output a hallucination probability score $s \in [0, 1]$ [1]. A single hidden layer MLP with ReLU non-linearity and sigmoid as the output activation to ouutput a score between 0 and 1.

4.3 Conditional Correction Steering

When s exceeds a preset threshold τ, a global correction vector $\hat{W}$ is dynamically applied to the output embeddings e_v using an adaptive steering signal α:

$$e'_v = e_v + \alpha \cdot \hat{W} \tag{2}$$

where $\hat{W}$ is the unit vector in the direction of:

$$W = \mu_{factual} - \mu_{hallucinated} \tag{3}$$

- $\mu_{factual}$ is the mean of all the feature maps of factual data observations at layer l^*.

- $\mu_{hallucinated}$ is the mean of all the feature maps of hallucinated data observations at layer l^*.

Unlike safety mechanisms that proceed to refusal, $\hat{W}$ steers away the model from hallucinations through LDA-inspired discriminative direction injection into hidden states

5 Conditional Linear Steering via Fisher Discriminant Direction

Let $h \in \mathbb{R}^d$ denote the hidden representation at a selected layer l^*. We consider two classes of representations obtained from paired prompts:

$$\mathcal{H}_f = \{h_i^{(f)}\}_{i=1}^N \quad \text{(factual)}, \qquad \mathcal{H}_h = \{h_i^{(h)}\}_{i=1}^N \quad \text{(hallucinated)}.$$

Consider the hallucination probable feature maps as a class and the factual feature maps as another class. Let the class means be,

$$\mu_f = \frac{1}{N}\sum_{i=1}^N h_i^{(f)}, \qquad \mu_h = \frac{1}{N}\sum_{i=1}^N h_i^{(h)}.$$

The classical Fisher Linear Discriminant direction is given by

$$w^* = S_W^{-1}(\mu_f - \mu_h),$$

where S_W is the within-class scatter matrix:

$$S_W = \sum_{i=1}^N (h_i^{(f)} - \mu_f)(h_i^{(f)} - \mu_f)^\top + \sum_{i=1}^N (h_i^{(h)} - \mu_h)(h_i^{(h)} - \mu_h)^\top.$$

In high-dimensional settings, we approximate S_W as $S_W \approx I$, leaving us with

$$w \approx \mu_f - \mu_h.$$

Thus, the steering vector used in our method is

$$W = \mu_f - \mu_h,$$

which serves as a discriminative direction separating factual and hallucinated representations.

Conditional activation steering. Let $s \in [0, 1]$ denote the hallucination probability score from a trained probe $\mathcal{P}$. We apply a threshold-based intervention:

$$h' = \begin{cases} h + \alpha W, & \text{if } s > \tau, \\ h, & \text{otherwise}, \end{cases}$$

where α is a fixed scaling hyperparameter and τ is a detection threshold.

Geometric interpretation. The update $h' = h + \alpha W$ translates the representation along the Fisher discriminant direction, thereby increasing its alignment with the factual class. Specifically,

$$W^\top h' = W^\top h + \alpha |W|^2,$$

Interpretation as linear decision boundary shift. In Fisher Linear Discriminant method, classification is done through the sign of $w^\top h$. The proposed framework increases $w^\top h$, effectively pushing representations toward the factual side of the decision boundary.

6 Experimental setup

- We have performed all our experiments on *llava-1.5-7b* model.
- All the experiments are run on *Nvidia GeForce RTX 3090 24GB* GPU.
- The Hallucination prober is a single hidden layer MLP with input dimension being *4096* and the hidden layer dimension as *512*. The prober is then trained using *Adam* optimizer and learning rate of 0.001 for 10 epochs.
- For the *llava-1.5-7b* model the l^* turned out to be layer 19.

7 Results and Conclusions

Method	Decoding	Acc	ΔAcc	Prec	ΔPrec	Recall	ΔRecall	F1	ΔF1
AutoSteer-H $\alpha = -2, \tau = 0.8$	Greedy	0.899	+0.0060	0.9716	+0.0757	0.8220	-0.0673	0.8906	-0.0020
	Beam	**0.912**	+0.0213	0.9654	+0.0450	0.8547	-0.0006	**0.9066**	+0.0199
	Nucleus	0.9023	+0.0429	0.9646	+0.1186	0.8353	-0.0435	0.8953	+0.0333
AutoSteer-H $\alpha = -1, \tau = 0.6$	Greedy	0.9013	+0.0083	0.9740	+0.0781	0.8247	-0.0646	0.8931	+0.0005
	Beam	0.8907	+0.0000	**0.9764**	+0.0560	0.8007	-0.0546	0.8799	-0.0068
	Nucleus	0.8943	+0.0349	0.9713	+0.1253	0.8127	-0.0661	0.8849	+0.0229
PAI	Greedy	0.893	–	0.8959	–	**0.8893**	–	0.8926	–
	Beam	0.8907	–	0.9204	–	0.8553	–	0.8867	–
	Nucleus	0.8594	–	0.8460	–	0.8788	–	0.8621	–

Table 1: Performance comparison of AutoSteer-H against PAI baseline. Δ denotes improvement over PAI for the same decoding strategy.

- **Precision and Accuracy Gains:** AutoSteer-H consistently outperforms PAI in precision (up to +12.53%) and accuracy (up to +4.29%) across all decoding strategies, demonstrating robust suppression of hallucinations.
- **F1 Performance:** We achieve a peak F1 score of **0.9066** (Beam, $\alpha = -2, \tau = 0.8$), with significant improvements under stochastic Nucleus sampling where baselines typically degrade.
- **Controlled Trade-offs:** A slight decrease in recall ranging from (-0.0673 to -0.0006) indicates a shift toward a strict factual output strategy. The threshold τ and steering strength α allow fine-tuning the balance between hallucination mitigation and general capability.
- **Improvements in Decoding Strategies:** Unlike PAI, whose performance decreases while using Nucleus sampling, but gains of AutoSteer-H remains stable.

Decoding	Method	CHAIR$_S$	CHAIR$_I$
Greedy	Vanilla	46.6	13.4
	PAI	24.8 (-21.8)	6.9 (-6.5)
	AutoSteer-H ($\alpha = -2, \tau = 0.8$)	22.4 (-24.2)	6.9 (-6.5)
	AutoSteer-H ($\alpha = -1, \tau = 0.6$)	22.2 (-24.4)	6.8 (-6.6)
Beam	Vanilla	46.4	14.3
	OPERA[5]	44.6 (-1.8)	14.4 (+0.1)
	PAI	21.8 (-24.6)	5.6 (-8.7)
	AutoSteer-H ($\alpha = -2, \tau = 0.8$)	20.2 (-26.2)	6.9 (-7.4)
	AutoSteer-H ($\alpha = -1, \tau = 0.6$)	19.6 (-26.8)	6.4 (-7.9)
Nucleus	Vanilla	58.2	18.2
	VCD[6]	51.8 (-6.4)	15.1 (-3.1)
	PAI	43.4 (-14.8)	14.7 (-3.5)
	AutoSteer-H ($\alpha = -2, \tau = 0.8$)	23.4 (-34.8)	8.4 (-9.8)
	AutoSteer-H ($\alpha = -1, \tau = 0.6$)	26.6 (-31.6)	8.9 (-9.3)

Table 2: CHAIR metrics comparison across frameworks and decoding strategies. Improvements relative to Vanilla decoding in each decoding strategy are shown.

- **Significant Hallucination Mitigation:** AutoSteer-H achieves the lowest CHAIR scores across all decoding methods, reducing sentence-level hallucinations ($CHAIR_S$) by up to 34.8% compared to the Vanilla model.
- **Improvements over Baselines:** Our method consistently outperforms specialized baselines like OPERA, VCD, and PAI. Notably, in Nucleus sampling, AutoSteer-H nearly halves the $CHAIR_S$ score compared to PAI (23.4 vs 43.4).
- **Object-Level Factuality:** The reduction in $CHAIR_I$ (instance-level) indicates that the steering vector effectively forces the model to ground its descriptions in actual visual features rather than relying on linguistic priors.

References

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